

# Lesson: Aspect Ratios

In mathematics, a **ratio** is a relationship between two numbers indicating how many times the first number contains the second. In film and television, Aspect Ratio is the ratio of the width to the height of the picture.

A ratio of 4:4 means that it is equal in size. A ratio of 4:3 means it is one unit wider than it is tall.

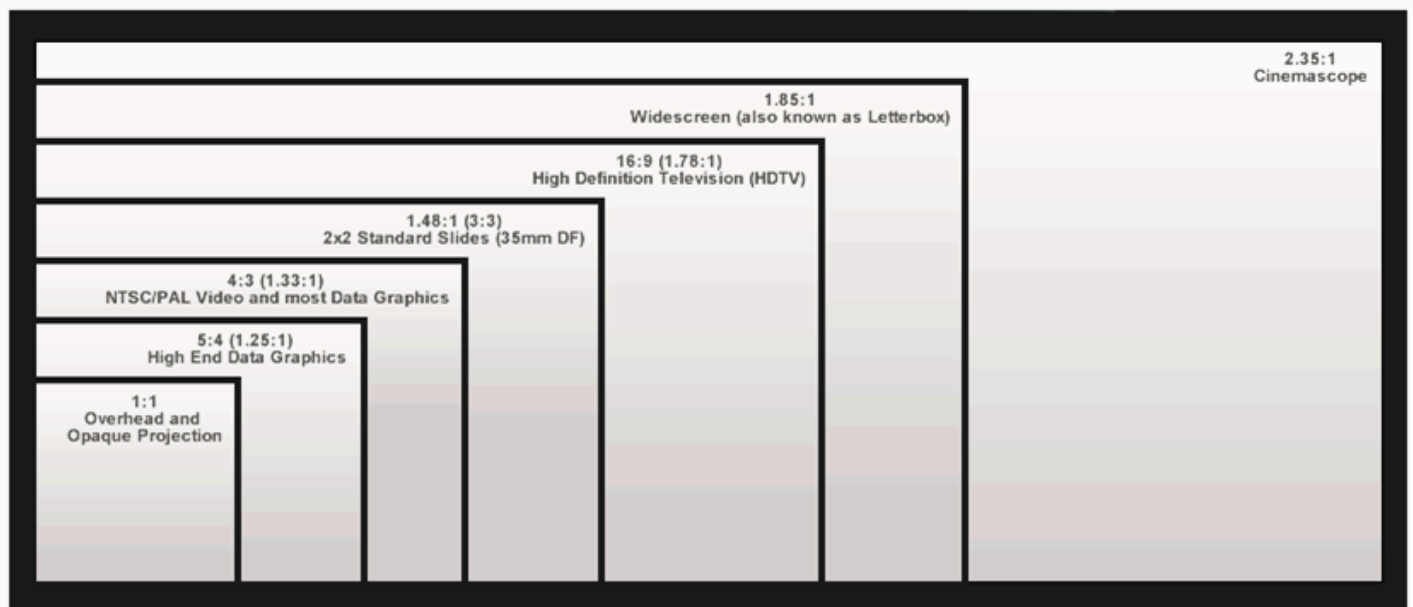
These are some common aspect ratios:

1.33:1 / 4:3 - Early television

1.78:1 / 16:9 - Modern television

1.85:1 - Standardized, nonanamorphic films (aka Letterboxed)

2.35:1 - Epic films



## Calculating Aspect Ratios

The formula for calculating aspect ratios for storyboards is **AR x Height = Width**

**Example:** A 1.85:1 aspect ratio with a panel height of 4 gives the panel width of 7.4 inches.  $1.85 \times 4 = 7.4$

You can also calculate the aspect ratio for a film or animation if you already know the width and height of the screen. Divide the screen width by its height to calculate the aspect ratio. **W / H = AR**

**Example:** A screen that is 4 inches wide by 3 inches tall will have a 1.33:1 aspect ratio.  $4 \div 3 = 1.33$

## Pan & Scan

Pan & Scan was a process that would crop the image to fit into a standard television format, focusing on the center of interest. This would only show part of the image and would have to be panned across to follow the action. Directors felt this was technically redirecting their films and the practice was subsequently discontinued. Today it is rare to find one of these films using this method.



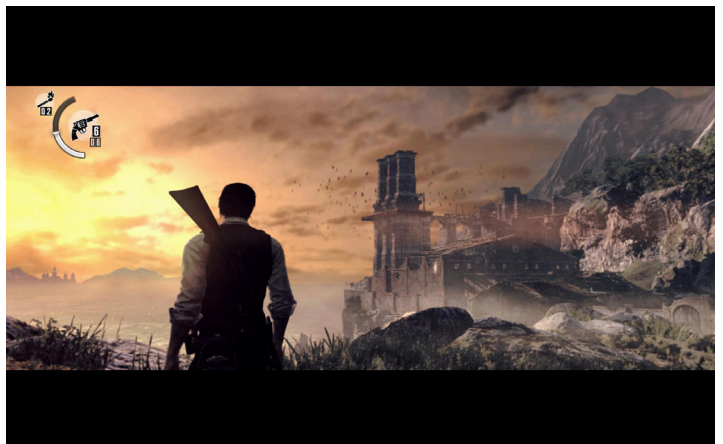
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### Letterboxing

Letterboxing is the process of fitting a widescreen image into a standard television screen by shrinking the image until it fits at its widest. This requires no calculations since the image is shrunk proportionately, resulting in black bars on top of and below the image. This process is adapted for films and video games.



The Evil Within

## The Good, The Bad, and The Ugly